Lin Li

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ACADEMIC APPOINTMENTS

Postdoctoral Researcher, University of Oxford 2025 – Present
Oxford Applied and Theoretical Machine Learning (OATML) Group, Department of CS, UK
- Advisor: Prof. Yarin Gal.
- Responsibilities: lead the projects on Fundational AI Safety; support group management and teaching activities; supervise MSc students; funding application.

Research Fellow, University of Birmingham 2025 – Present
School of Computer Science, Birmingham, UK
- Working with: Prof. Huiping Chen.
- Responsibilities: co-lead the project; direct the research; co-supervise MSc students.

Research Intern, Tencent 2021 – 22
Robotics X Lab, Shenzhen, China
- Mentor: Dr. Lipeng Chen.
- Responsibilities: co-led the project on enabling robots to learn throwing and catching skills.

EDUCATION

PhD in Machine Learning, King's College London 2019 – 24
Department of Informatics, London, UK
- Advisors: Prof. Michael Spratling (primary) and Prof. Dimitrios Letsios.
- Thesis: *Towards Robust Visual Classification through Adversarial Training.*
- Thesis Committee: Prof. George D. Magoulas and Dr. Adel Bibi.

MSc in Computing Science, Imperial College London 2017 – 18
Department of Computing, London, UK
- Advisor: Prof. Wayne Luk.
- Grade: Overall Distinction (Exam + Thesis).
- Dissertation: *Understanding Deep CNNs via Interpretable Individual Units*.

BM in Financial Management, Xiamen University 2013 – 17
Department of Finance, School of Management, Xiamen, China
- Advisor: Prof. Zheng Qiao.
- Grade: top 10% of graduating class.
- Dissertation: *Quantitatively Measuring Investor's Sentiment via Search Index.*

PUBLICATIONS

(* indicates equal contribution; † indicates the corresponding author if have)

AI Safety:

1. **Lin Li**†, Michael Spratling, "Robust Shortcut and Disordered Robustness: Improving Adversarial Training Through Adaptive Smoothing", Pattern Recognition (**PR**), 2025
2. **Lin Li**, Yifei Wang, Chawin Sitawarin, Michael Spratling, "OODRobustBench: a Benchmark and Large-Scale Analysis of Adversarial Robustness under Distribution Shift", International Conference on Machine Learning (**ICML**) 2024 and ICLR workshop Data-centric Machine Learning Research, 2024
3. **Lin Li***, Haoyan Guan*, Jianing Qiu, Michael Spratling, "One Prompt Word is Enough to Boost Adversarial Robustness for Pre-trained Vision-Language Models", IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR**), 2024
4. **Lin Li**†, Jianing Qiu, Michael Spratling, "AROID: Improving Adversarial Robustness Through Online Instance-Wise Data Augmentation", the International Journal of Computer Vision (**IJCV**), 2024
5. **Lin Li**, Michael Spratling, "Data augmentation alone can improve adversarial training", International Conference on Learning Representations (**ICLR**), 2023
6. **Lin Li**, Michael Spratling†, "Understanding and combating robust overfitting via input loss landscape analysis and regularization", Pattern Recognition (**PR**), 2023
7. Jahan C Penny-Dimri, Magdalena Bachmann, William R Cooke, Sam Mathewlynn, Samuel Dockree, John Tolladay, Jannik Kossen, **Lin Li**, Yarin Gal, Gabriel Davis Jones, "Reducing Large Language Model Safety Risks in Women's Health using Semantic Entropy", in submission, 2025
8. Jianing Qiu, **Lin Li**, Jiankai Sun, Hao Wei, Zhe Xu, Kyle Lam, Wu Yuan, "Emerging Cyber Attack Risks of Medical AI Agents", in submission, 2025

AI + Healthcare / Robotics:

9. Jianing Qiu, **Lin Li**, Jiankai Sun, and Jiachuan Peng, Peilun Shi, Ruiyang Zhang, Yinzhao Dong, Kyle Lam, Frank P.-W. Lo, Bo Xiao, Wu Yuan, Dong Xu†, Benny Lo†, "Large AI Models in Health Informatics: Applications, Challenges, and the Future", IEEE Journal of Biomedical and Health Informatics (**JBHI**), 2023
10. Lipeng Chen*†, Jianing Qiu*, **Lin Li***, Xi Luo, Guoyi Chi, Yu Zheng, "Advancing Robots with Greater Dynamic Dexterity: A Large-Scale Multi-View and Multi-Modal Dataset of Human-Human Throw&Catch of Arbitrary Objects", International Journal of Robotics Research (**IJRR**), 2024
11. Jianing Qiu*, Jian Wu*, Hao Wei*, and Peilun Shi, Minqing Zhang, Yunyun Sun, **Lin Li**, … (+34 authors), Wu Yuan†, "Development and Validation of a Multimodal Multitask Vision Foundation Model for Generalist Ophthalmic Artificial Intelligence", New England Journal of Medicine Artificial Intelligence (**NEJM AI**), 2024
12. Linyuan Li, Anujit Saha, **Lin Li**, Boyuan Li, Mengxian He, Jianing Qiu, Wu Yuan, "Artificial Intelligence for Biomedical Video Generatio", in submission, 2024

SOFTWARES & OPEN-SOURCE

OODRobustBench 2024

- Description: the first benchmark for evaluating adversarial robustness under distribution shifts.
- Contribution: led the project; developed the concept, code, and leaderboards.
- Leaderboards, Evaluation Toolbox.

Human-Human Throw&Catch (H2TC) Dataset 2024

- Description: the first large multimodal dataset of human throw-catch actions for humanoid robots.
- Contribution: co-led the project; co-developed the concept, code, and dataset.
- Website (Data, Tools and Tutorials).

Awesome-Healthcare-Foundation-Model 2023

- Description: a continual paper collection of foundation models for healthcare (46 folks and 450 stars).
- Contribution: co-created the collection; co-authored a survey based on it.
- Paper collection.

Adversarial Robustness Toolbox (ART), Linux Foundation AI & Data Foundation 2020

- Description: a popular (1.2k+ folks and 5.2k+ stars on Github) library for adversarial ML research.
- Contribution: fix the bugs as recognized here.

Smart Acc, Microsoft 2018

- Description: a web application based on object detection algorithms for accomodation search.
- Contribution: co-led the project; developed the core object detection algorithms.
- Blog (Microsoft), Code, Report, Demo.

GRANTS

Horizon Europe (€29M) 2025-27

Proposal Contributor, Postdoctoral Researcher

- Description: the EU's key funding programme for research and innovation.
- Proposal Title: Diversis Viis, Plurima Solvens (DVPS)
- Responsibilities: co-developed the concept of multimodal foundation models for human and society; wrote the corresponding sections in the proposal; lead research on multimodal hallucination.

IDAI Pump Prime (£20K) 2025

Key Personnel (named on the proposal)

- Description: a priming fund from The Institute for Data and AI (IDAI), University of Birmingham to support the development of innovative, interdisciplinary and high-impact research ideas.
- Proposal Title: Proactive defense against Deepfake
- Responsibilities: co-lead the project; direct the research; co-supervise MSc students.

AWARDS & HONORS

PGR Research Support (£2.5K), KCL 2023

King's-China Scholarship (Tuition waiver + £16.2K per year), KCL and CSC 2019-23

1st Class (Xiangyu) University Scholarship (¥7K), Xiamen University 2016

Excellent Academic Performance Scholarship (¥1K), Xiamen University 2015

3rd prize winner (¥5K), Jinyuan Creativity and Startup Contest, Xiamen City 2016

3rd prize winner, ChinaNet Dream Accelerator Programming Contest, China 2015

TEACHING

Teaching Assistant 2025

Department of Computer Science, University of Oxford, Oxford, UK

- Course: Uncertainty in Deep Learning (Graduate).
- Responsibilities: marking individual projects.

Teaching Assistant 2021

Department of Informatics, King's College London, London, UK

- Courses: Machine Learning and Pattern Recognition (Graduate), Introduction to Artificial Intelligence (UG).
- Responsibilities: small group tutorial, lab sessions, marking courseworks. I also co-developed the automated marking tool for individual projects in the Machine Learning and Pattern Recognition course, which remains in use to this day.

INVITED TALKS

CHI Seminar 2025

Lab internal seminar, Computational Health Informatics (CHI) Lab, University of Oxford, UK

- Title: Prompting VLMs for accuracy and robustness

Nanqiang Youth Scholar Forum 2024

Internatioanl faculty recruitment event, Xiamen University, China.

- Title: Towards Safe Multimodal Foundational Models in Finance.

Vision And Learning Seminar (VALSE) 2024

National ML/CV seminar (5k+ attendees), China.

- Title: One Prompt Word is Enough to Boost Adversarial Robustness for Pre-trained VLMs.

AI Time Youth PhD Talk 2023

Online AI forum, Tsinghua University, China.

- Title: Data augmentation can improve adversarial training.

Departmental Research Showcase 2023

Research showcase to the public, KCL, UK.

- Title: Defending DNNs against adversarial examples.

ACADEMIC SERVICE

Reviewer

- Conferences: NeurIPS, ICML, ICLR, AAAI, ECCV.
- Journals: IEEE T-Pattern Analysis and Machine Intelligence (T-PAMI), Pattern Recognition (PR), IEEE T-Information Forensics & Security (T-IFS), IEEE T-Dependable and Secure Computing (T-DSC), IEEE Journal of Biomedical and Health Informatics (J-BHI).

Program Committee

- Workshops:
  1. 2nd MEIS Workshop@CVPR2025
  2. Workshop on Cooperative Intelligence for Embodied AI@ECCV2024
  3. WIHR@ICRA 2024

Organizing Committee

- Challenge: Human–Robot Object Throwing and Catching Challenge@CVPR2025

ACADEMIC REFEREES

(refer to their websites for the latest information)

Yarin Gal

Associate Professor of ML, Department of Computer Science, University of Oxford.
Turing AI Fellow, The Alan Turing Institute, London, UK.
Director of Research, UK AI Security Institute (AISI), London, UK.
Tutorial Fellow, Christ Church College, University of Oxford.
Phone: +44 1865 610497, Email: yarin@cs.ox.ac.uk.
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Michael Spratling

Researcher, Department of Behavioural and Cognitive Sciences, University of Luxembourg.
was Reader[1] in Computational Neuroscience and Visual Cognition, Department of Informatics, KCL.
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[1] Michael held this position while serving as my PhD supervisor.

Adel Bibi
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